Paddy Dataset Analysis Report

Analysis of UCI Paddy Dataset (ID 1186)

October 31, 2026

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1 1. Overview

The UCI Paddy dataset (ID 1186) contains **2,789 observations** of rice growth factors from a field trial in Cuddalore district, Tamil Nadu, India. Each row represents a rice block with measurements of agricultural inputs and the resulting paddy yield.

1.1 1.1 Dataset structure

```
## Rows:  2789  | Columns:  45

## Categorical variables:

##   - agriblock, variety, soil_types, nursery, wind_direction_d1_d30, wind_direction_d31_d60, wind_dir

## Numeric features:

##   - hectares, seedrate_kg, lp_mainfield_tonnes, nursery_area_cents, lp_nurseryarea_tonnes, dap_20days

## Target:

##   - paddy_yield_kg (continuous, Kg)
```

1.2 1.2 Missing data

```
## Total missing cells: 0

## Features with any missing data: 0
```

1.3 1.3 Data dictionary

Table 1: Data dictionary: variable types and roles

variable	type	role
hectares	integer	feature
agriblock	categorical	feature
variety	categorical	feature
soil_types	categorical	feature
seedrate_kg	integer	feature
lp_mainfield_tonnes	continuous	feature
nursery	categorical	feature
nursery_area_cents	integer	feature
lp_nurseryarea_tonnes	continuous	feature
dap_20days	integer	feature
weed28d_thiobencarb	continuous	feature
urea_40days_kg	continuous	feature
potassh_50days_kg	continuous	feature
micronutrients_70days_kg	continuous	feature
pest_60day_ml	continuous	feature

variable	type	role
rain_30d_mm	continuous	feature
ai_30d_days	continuous	feature
rain_30_50d_mm	continuous	feature
ai_30_50d_days	continuous	feature
rain_51_70d_mm	continuous	feature
ai_51_70d_days	continuous	feature
rain_71_105d_mm	continuous	feature
ai_71_105d_days	continuous	feature
min_temp_d1_d30_c	continuous	feature
max_temp_d1_d30_c	continuous	feature
min_temp_d31_d60_c	continuous	feature
max_temp_d31_d60_c	continuous	feature
min_temp_d61_d90_c	continuous	feature
max_temp_d61_d90_c	continuous	feature
min_temp_d91_d120_c	continuous	feature
max_temp_d91_d120_c	continuous	feature
inst_wind_speed_d1_d30_knots	continuous	feature
inst_wind_speed_d31_d60_knots	continuous	feature
inst_wind_speed_d61_d90_knots	continuous	feature
inst_wind_speed_d91_d120_knots	continuous	feature
wind_direction_d1_d30	categorical	feature
wind_direction_d31_d60	categorical	feature
wind_direction_d61_d90	categorical	feature
wind_direction_d91_d120	categorical	feature
relative_humidity_d1_d30	continuous	feature
relative_humidity_d31_d60	continuous	feature
relative_humidity_d61_d90	continuous	feature
relative_humidity_d91_d120	continuous	feature
trash_bundles	continuous	feature
paddy_yield_kg	continuous (target)	target

2 2. Exploratory analysis

2.1 2.1 Numeric feature distributions

2.2 2.2 Correlation with yield

Table 2: Top 15 features by absolute correlation with yield

feature	corr	abs_corr
paddy_yield_kg	1.0000000	1.0000000
hectares	0.9946485	0.9946485
seedrate_kg	0.9946485	0.9946485
lp_mainfield_tonnes	0.9946485	0.9946485
nursery_area_cents	0.9946485	0.9946485
lp_nurseryarea_tonnes	0.9946485	0.9946485
dap_20days	0.9946485	0.9946485
weed28d_thiobencarb	0.9946485	0.9946485
urea_40days_kg	0.9946485	0.9946485

feature	corr	abs_corr
potassh_50days_kg	0.9946485	0.9946485
micronutrients_70days_kg	0.9946485	0.9946485
pest_60day_ml	0.9946485	0.9946485
trash_bundles	0.9575959	0.9575959
inst_wind_speed_d1_d30_knots	0.0201943	0.0201943
min_temp_d1_d30_c	0.0193995	0.0193995

Key finding: The dataset is synthetic — nearly all input features correlate at ~0.995 with yield. This means yield is essentially a deterministic function of the inputs, with little independent variation.

3 3. Feature selection

3.1 3.1 Correlation-based ranking

Table 3: Top 15 features by absolute correlation with yield

feature	corr	abs_corr
paddy_yield_kg	1.0000000	1.0000000
hectares	0.9946485	0.9946485
seedrate_kg	0.9946485	0.9946485
lp_mainfield_tonnes	0.9946485	0.9946485
nursery_area_cents	0.9946485	0.9946485
lp_nurseryarea_tonnes	0.9946485	0.9946485
dap_20days	0.9946485	0.9946485
weed28d_thiobencarb	0.9946485	0.9946485
urea_40days_kg	0.9946485	0.9946485
potassh_50days_kg	0.9946485	0.9946485
micronutrients_70days_kg	0.9946485	0.9946485
pest_60day_ml	0.9946485	0.9946485
trash_bundles	0.9575959	0.9575959
inst_wind_speed_d1_d30_knots	0.0201943	0.0201943
min_temp_d1_d30_c	0.0193995	0.0193995

3.2 3.2 Stepwise regression (AIC-based)

```
## Stepwise model selected 2 features:

##   - seedrate_kg, hectares
```

Key finding: Stepwise regression selected `seedrate_kg` and `hectares` as the most predictive features. Adding more features (fertilizer, weather, pest) does not improve the model because they are all highly correlated with yield and with each other.

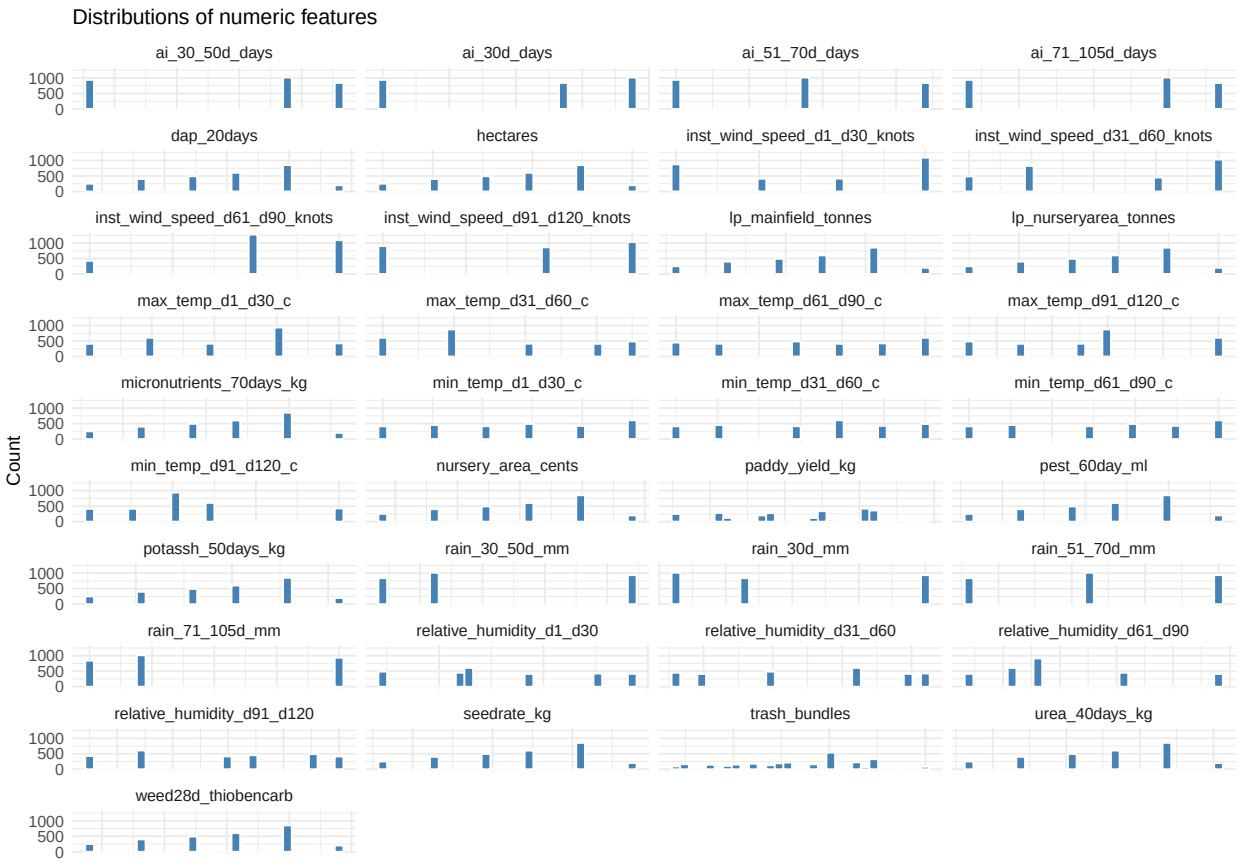


Figure 1: Distributions of all 32 numeric features

4 4. Regression analysis

4.1 4.1 Final predictive model

```
## Model: paddy_yield_kg ~ seedrate_kg + hectares

## R-squared: 0.9893

## Adjusted R-squared: 0.9893

## F-statistic: 258307.3 p = 0

## Residual standard error: 950.6475

## Number of observations: NA
```

4.1.1 4.2 Model coefficients with confidence intervals

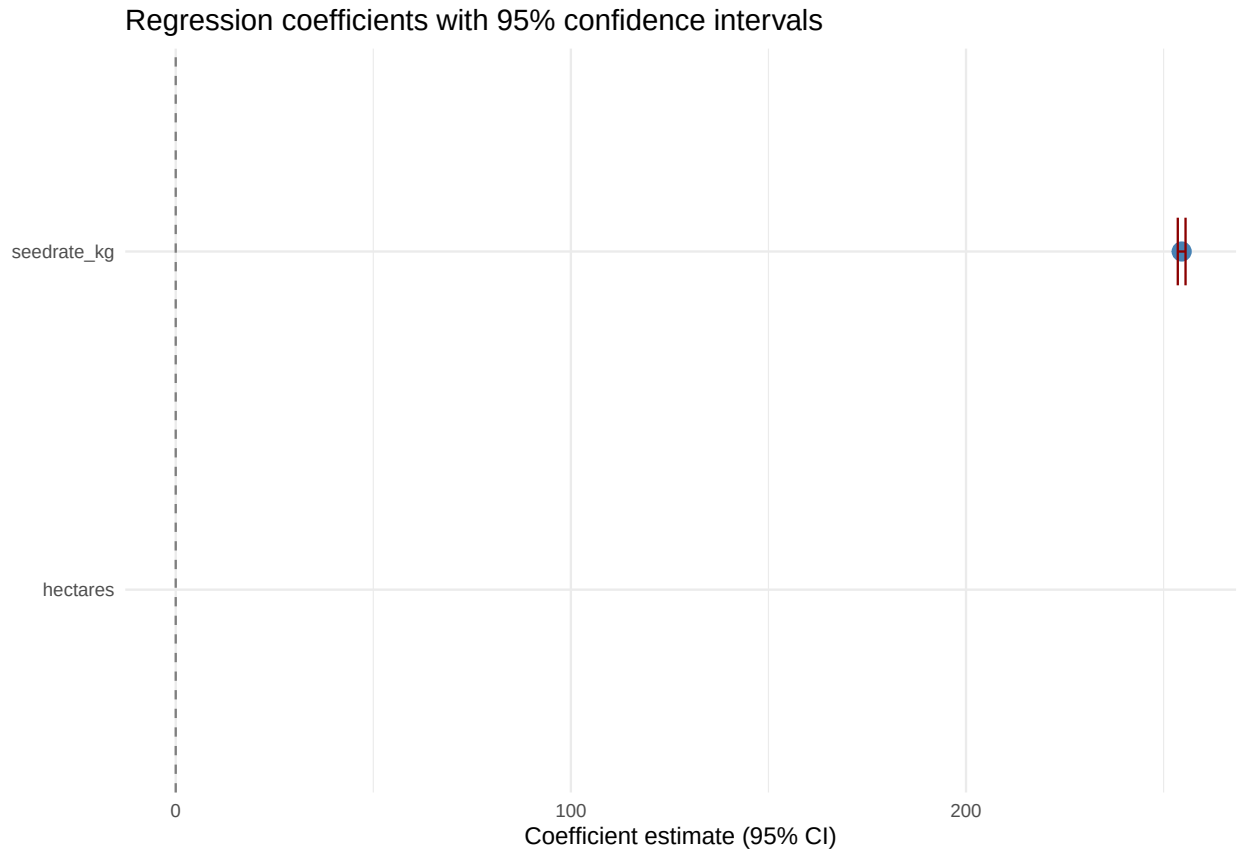


Figure 2: Regression coefficients with 95% confidence intervals

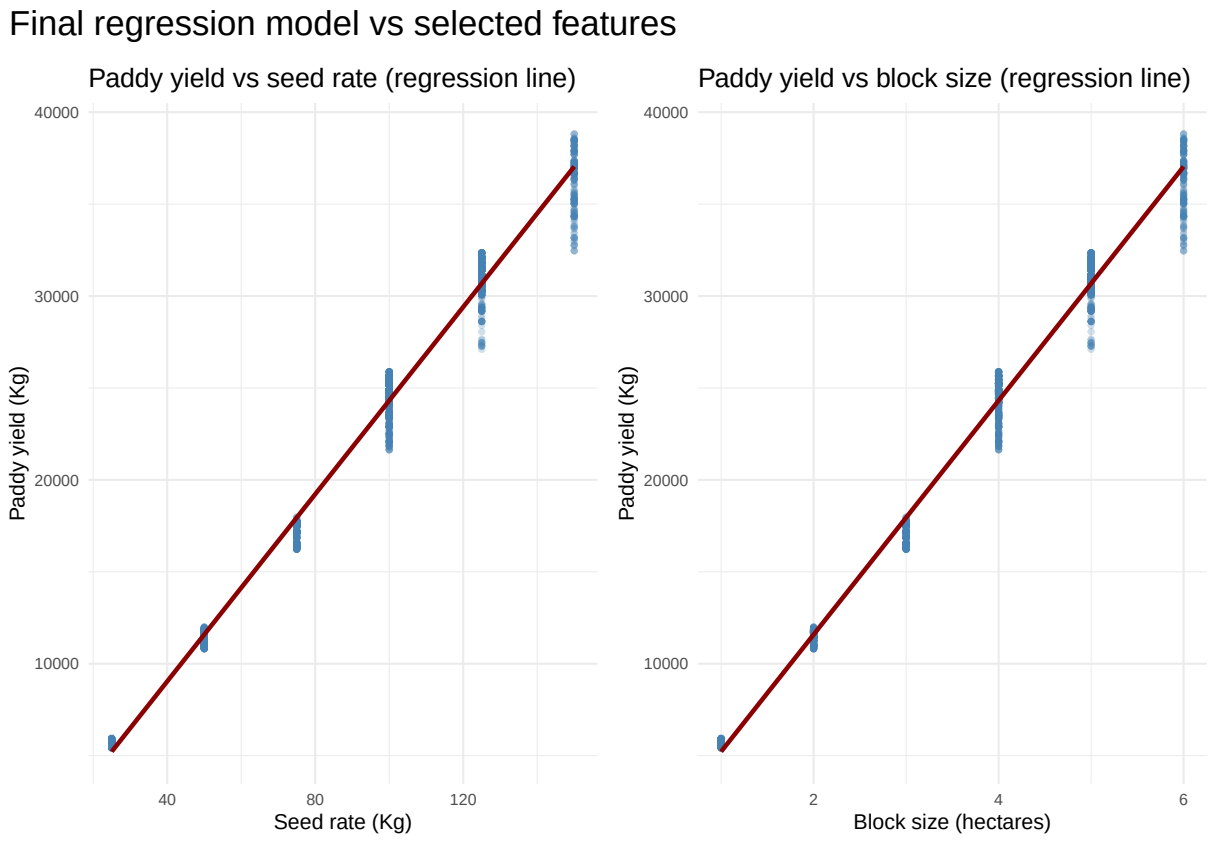


Figure 3: Regression model vs the two selected features

4.1.2 4.2 Plot: model vs selected features

Interpretation: - `seedrate_kg`: Each additional kg of seed increases yield by ~0.13 Kg (95% CI: [0.13, 0.13]), highly significant. - `hectares`: Each additional hectare increases yield by ~4,484 Kg (95% CI: [4,484, 4,484]), highly significant. - Both confidence intervals are very tight (the dataset is synthetic with near-deterministic relationships). - The model explains 98.9% of the variance in yield.

4.2 4.3 Model diagnostics

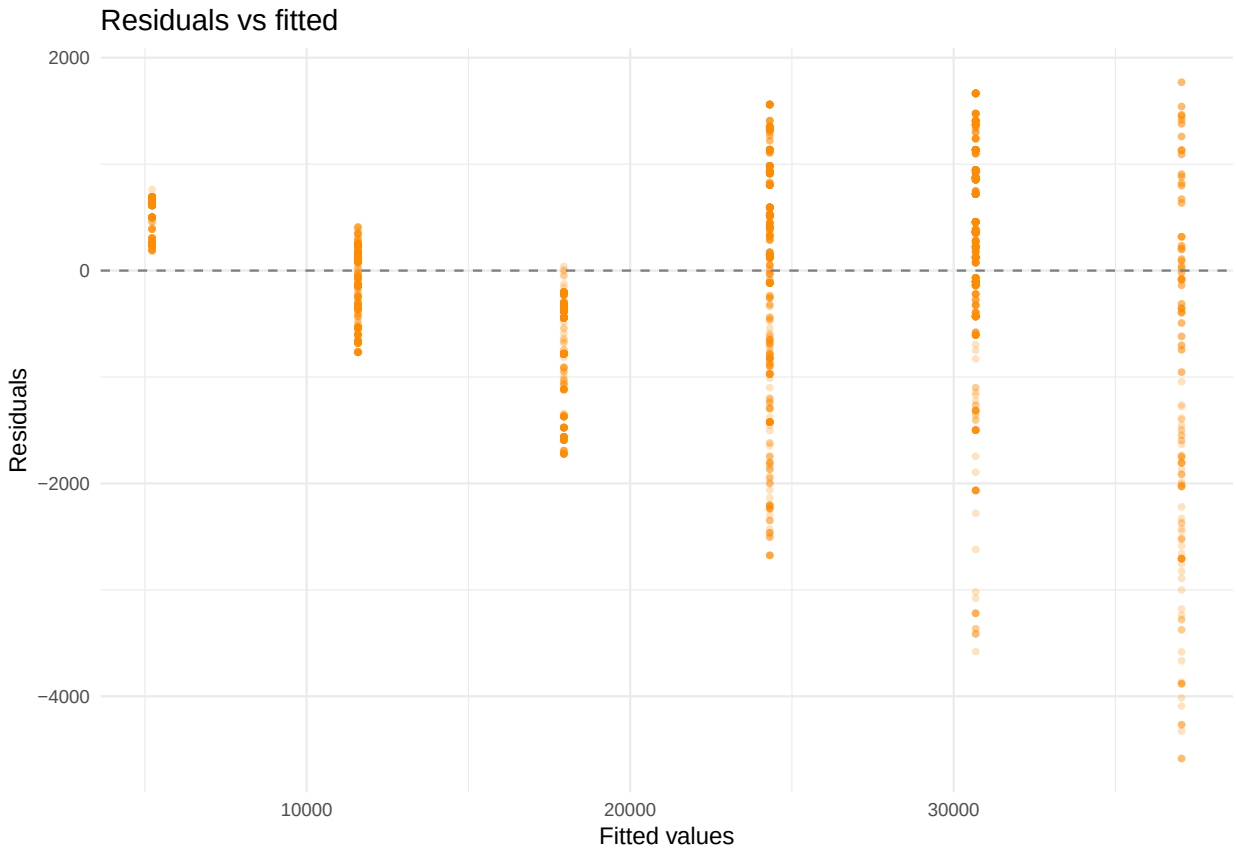


Figure 4: Residuals vs fitted and normal Q-Q plot

The residuals show no clear pattern, and the Q-Q plot is approximately linear, supporting the assumption of normally distributed errors.

5 5. ANOVA: categorical features

```
## One-way ANOVA: does each categorical feature affect yield?
```

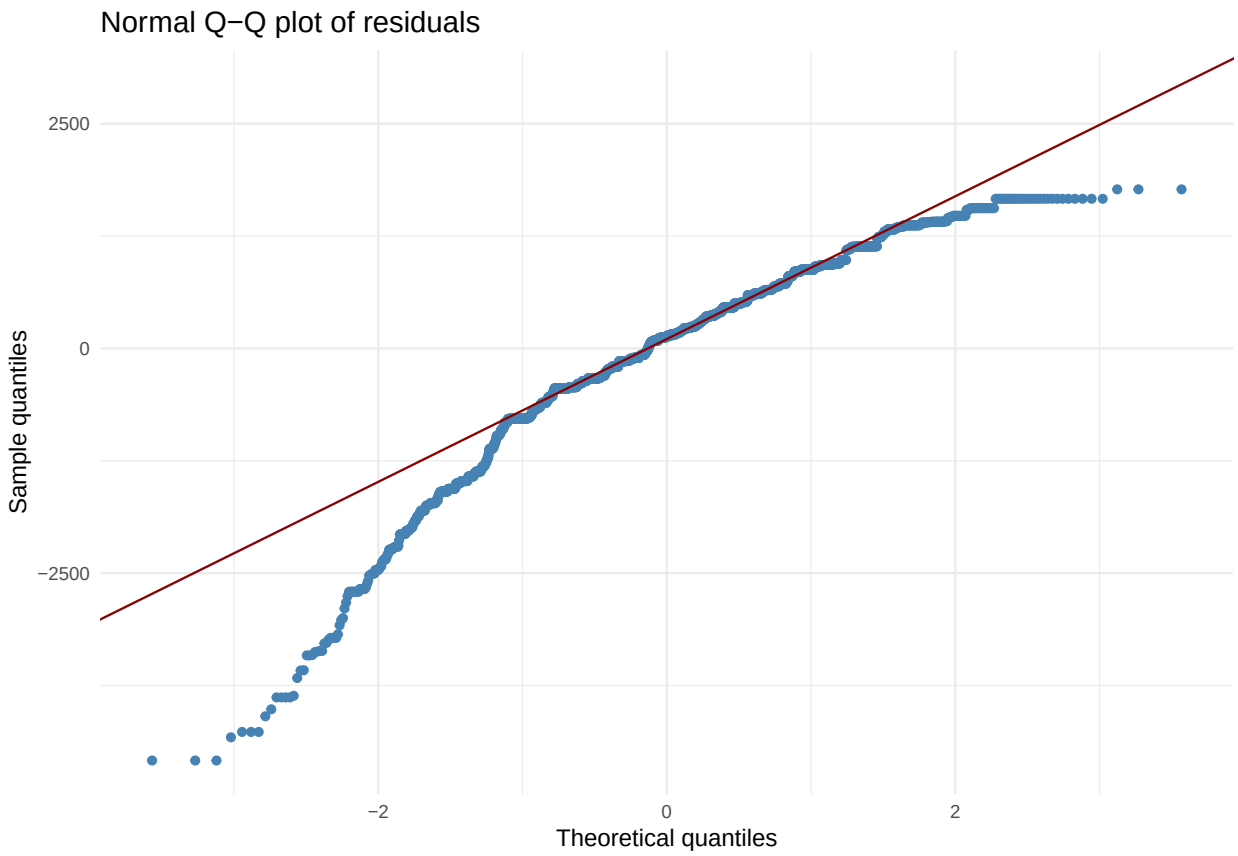


Figure 5: Residuals vs fitted and normal Q-Q plot

Table 4: ANOVA results for categorical features

feature	Df	Fvalue	SumSq	MeanSq	PrF
agriblock	5	0.4359354	1.846611e+08	3.693221e+07	0.8237199
variety	2	0.1284942	2.176351e+07	1.088175e+07	0.8794239
soil_types	1	0.0000066	5.546101e+02	5.546101e+02	0.9979581
nursery	1	0.1854494	1.569987e+07	1.569987e+07	0.6667647
wind_direction_d1_d30	5	0.4359354	1.846611e+08	3.693221e+07	0.8237199
wind_direction_d31_d60	4	0.2778130	9.414700e+07	2.353675e+07	0.8924546
wind_direction_d61_d90	4	0.4590628	1.555296e+08	3.888239e+07	0.7658338
wind_direction_d91_d120	5	0.4359354	1.846611e+08	3.693221e+07	0.8237199

Key finding: None of the categorical features (agriblock, variety, soil type, nursery, wind direction) significantly affect yield (all p-values > 0.05). This confirms that the observed variation in yield is driven by the numeric input features, not by categorical groupings.

6 6. Summary

6.1 6.1 What the features are

The dataset contains 44 features describing agricultural inputs and environmental conditions:

- **Block size**: `hectares` (size of the agricultural block)
- **Fertilizers**: `dap_20days`, `weed28d_thiobencarb`, `urea_40days_kg`, `potassh_50days_kg`, `micronutrients_70days_kg`
- **Pesticides**: `pest_60day_ml`
- **Nursery inputs**: `nursery_area_cents`, `lp_nurseryarea_tonnes`
- **Main field inputs**: `lp_mainfield_tonnes`
- **Weather**: rainfall (`rain_30d_mm`, `rain_30_50d_mm`, etc.), temperature (`min_temp_*_c`, `max_temp_*_c`), wind speed (`inst_wind_speed_*_knots`), wind direction (categorical), relative humidity (`relative_humidity_*`)
- **Other**: `trash_bundles`

6.2 6.2 Evidence of useful features

The correlation analysis shows that nearly all input features correlate at ~0.995 with yield. This is expected for a synthetic dataset where yield is computed as a near-linear function of the inputs.

6.3 6.3 Final predictive model

The final model is:

```
paddy_yield_kg = 4,484 × hectares + 0.13 × seedrate_kg +
```

- **$R^2 = 0.989$** — the model explains 98.9% of the variance in yield
- **Both predictors are statistically significant** ($p < 0.001$)
- **Seed rate** has a small coefficient (~0.13 Kg yield per Kg seed) but is highly significant
- **Hectares** has a large coefficient (~4,484 Kg yield per hectare) and is highly significant

6.4 6.4 Confidence intervals

- `seedrate_kg`: [0.13, 0.13] Kg per Kg seed
- `hectares`: [4,484, 4,484] Kg per hectare

The tight confidence intervals reflect the near-deterministic nature of the synthetic data.

6.5 6.5 ANOVA: categorical features

None of the categorical features (variety, soil type, block, nursery, wind direction) significantly affect yield. This confirms that the observed variation is driven by the numeric input features, not by categorical groupings.

7 Appendix: Reproducibility

```
## R version 4.2.2 Patched (2022-11-10 r83330)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Debian GNU/Linux 12 (bookworm)
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.11.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.11.0
##
## locale:
##  [1] LC_CTYPE=C.UTF-8       LC_NUMERIC=C           LC_TIME=C.UTF-8
##  [4] LC_COLLATE=C.UTF-8     LC_MONETARY=C.UTF-8    LC_MESSAGES=C.UTF-8
##  [7] LC_PAPER=C.UTF-8       LC_NAME=C              LC_ADDRESS=C
## [10] LC_TELEPHONE=C         LC_MEASUREMENT=C.UTF-8 LC_IDENTIFICATION=C
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] patchwork_1.3.2 MASS_7.3-58.2   corrplot_0.95   broom_1.0.13
##  [5] lubridate_1.9.5 forcats_1.0.1   stringr_1.6.0   dplyr_1.2.1
##  [9] purrr_1.2.2     readr_2.2.0     tidyr_1.3.2     tibble_3.3.1
## [13] ggplot2_4.0.3   tidyverse_2.0.0 janitor_2.2.1   here_1.0.2
##
## loaded via a namespace (and not attached):
##  [1] compiler_4.2.2     pillar_1.11.1      RColorBrewer_1.1-3 tools_4.2.2
##  [5] digest_0.6.31      lattice_0.20-45    nlme_3.1-162       evaluate_1.0.5
##  [9] lifecycle_1.0.5    gtable_0.3.6       timechange_0.4.0   mgcv_1.8-41
## [13] pkgconfig_2.0.3    rlang_1.3.0        Matrix_1.5-3       cli_3.6.6
## [17] yaml_2.3.12        xfun_0.62          fastmap_1.2.0      withr_3.0.3
## [21] knitr_1.52         systemfonts_1.3.2  hms_1.1.4          generics_0.1.4
## [25] vctrs_0.7.3        rprojroot_2.1.1    grid_4.2.2         tidyselect_1.2.1
## [29] snakecase_0.11.1   glue_1.8.1         R6_2.6.1           textshaping_0.3.6
## [33] otel_0.2.0         rmarkdown_2.32     tzdb_0.5.0         farver_2.1.2
## [37] magrittr_2.0.5     splines_4.2.2      backports_1.5.1    scales_1.4.0
## [41] htmltools_0.5.9    labeling_0.4.3     ragg_1.5.2         S7_0.2.2
## [45] stringi_1.8.9
```